

AI-Enabled Early Breast Cancer Detection Platform

Financial Impact Analysis for Pakistani Healthcare System

Prepared: January 2026

IMPORTANT DISCLAIMER: This document presents a framework for financial analysis based on available literature and publicly accessible information. Cost figures marked with (*) require validation through primary research with Pakistani healthcare institutions. All projections are estimates pending actual pilot program implementation.

Executive Summary

This analysis presents evidence-based financial projections for implementing AI-powered breast cancer risk assessment in Pakistan's healthcare system. The platform demonstrates significant potential for cost savings through early detection, reduced false positives, and operational efficiency improvements. Conservative estimates project PKR 200-500 million in annual value per mid-sized hospital.

1. Pakistani Healthcare Context

Current Breast Cancer Burden

Metric	Value	Source Status
Breast Cancer Incidence	Highest in South Asia	✓ Verified
Late-stage Detection Rate	90% of cases	■ Needs Primary Verification
5-year Survival Rate	30-40% vs 85%+ in developed countries	✓ International Literature
Healthcare Access	Limited screening infrastructure	✓ Government Reports

Verified International Sources:

- GLOBOCAN 2020** - International Agency for Research on Cancer (IARC): Breast cancer statistics for Pakistan
- World Health Organization** - Global Health Observatory data repository
- McKinney et al. (2020)** - "International evaluation of an AI system for breast cancer screening" - Nature
- Rodriguez-Ruiz et al. (2019)** - "Stand-alone artificial intelligence for breast cancer detection" - Radiology

2. Treatment Cost Analysis (PKR, 2024)

Cost Estimates Based on Available Data (*Requires Validation)

Treatment Stage	Cost Range (PKR)*	Source Required	Confidence Level
Early Stage (I-II)	300,000 - 800,000	AKU/SKMCH Cost Studies	Medium
Late Stage (III-IV)	1.2M - 3.5M	Hospital Financial Data	Medium
Cost Differential	3-4x Higher for Late Stage	International Literature	High

Required Primary Research Sources:

Essential Data Collection Needed:

- Aga Khan University Hospital - Treatment cost analysis
- Shaukat Khanum Memorial Cancer Hospital - Patient cost studies
- Combined Military Hospital (CMH) - Public sector costs
- Pakistan Health Research Council - National cost database

- Ministry of National Health Services - Policy cost frameworks

3. Imaging and Diagnostic Costs

Market Rate Estimates (*Primary Research Required)

Procedure	Public Hospital*	Private Hospital*	Research Status
Screening Mammography	PKR 2,500-4,000	PKR 5,000-8,000	Needs Survey
Diagnostic Mammography	PKR 3,500-5,500	PKR 6,000-10,000	Needs Survey
Breast Ultrasound	PKR 2,000-3,500	PKR 4,000-7,000	Needs Survey
Breast MRI	PKR 18,000-25,000	PKR 30,000-50,000	Needs Survey

4. Conservative Financial Impact Projections

Scenario Analysis (Annual, 10,000 screenings)

Methodology: Based on verified international AI screening studies showing 2-8% improvements in early detection rates, applied to Pakistani healthcare context using conservative assumptions.

Scenario	Early Detection Improvement	Additional Cases	Annual Savings (PKR)
Conservative	2%	200 cases	PKR 180 million
Moderate	5%	500 cases	PKR 450 million
Optimistic	8%	800 cases	PKR 720 million

5. Total Economic Impact Summary

Annual Benefits Per Hospital (10,000 screenings)

Benefit Category	Conservative	Moderate	Optimistic
Early Detection Savings	PKR 180M	PKR 450M	PKR 720M
Reduced False Positives	PKR 2M	PKR 3M	PKR 5M
Operational Efficiency	PKR 4M	PKR 6M	PKR 8M
Increased Throughput	PKR 3M	PKR 5M	PKR 8M
TOTAL ANNUAL VALUE	PKR 189M	PKR 464M	PKR 741M

RECOMMENDED PLANNING FIGURE: PKR 200-500 Million Annual Value per Hospital

This conservative range accounts for implementation challenges and requires validation through pilot programs.

6. Implementation Pathway

Phase	Timeline	Activities	Investment (PKR)	Expected ROI
Phase 1: Pilot	Months 1-6	Deploy at AKU & SKMCH Validate assumptions	15-25 million	Data validation
Phase 2: Scale	Months 7-18	Deploy across 10 hospitals 100K+ screenings	50-100 million	100-200M revenue
Phase 3: National	Years 2-5	National Health integration 1M+ screenings	200-500 million	1-2B revenue

7. Required Research and Validation

Critical Data Collection Priorities

Immediate Research Requirements (Month 1-3):

- 1. Treatment Cost Validation** - Partner with 5+ major hospitals for actual cost data
- 2. Current Screening Volumes** - Survey existing mammography capacity and utilization
- 3. Detection Rate Baselines** - Establish current early vs. late-stage detection percentages
- 4. Healthcare Workforce Assessment** - Evaluate radiologist capacity and workflow constraints

Partner Institutions for Data Validation:

- Aga Khan University Hospital (Karachi)
- Shaukat Khanum Memorial Cancer Hospital (Lahore)
- Combined Military Hospital (Rawalpindi)
- Pakistan Health Research Council
- Pakistan Society of Radiology
- Ministry of National Health Services

Risk Mitigation Strategy:

- Begin with 6-month pilot at 2 major hospitals
- Validate all cost assumptions before full deployment
- Adjust financial projections based on actual performance data
- Establish baseline metrics during pilot phase

8. References and Source Requirements

Verified International References:

1. McKinney, S.M., et al. (2020). "International evaluation of an AI system for breast cancer screening." *Nature*, 577(7788), 89-94. DOI: 10.1038/s41586-019-1799-6
2. Rodriguez-Ruiz, A., et al. (2019). "Stand-alone artificial intelligence for breast cancer detection in mammography." *Radiology*, 293(3), 562-571. DOI: 10.1148/radiol.2019191370
3. GLOBOCAN 2020. International Agency for Research on Cancer (IARC). "Cancer statistics for Pakistan." Available: <https://gco.iarc.fr/today/data/factsheets/populations/586-pakistan-fact-sheets.pdf>
4. World Health Organization (2023). "Global Health Observatory data repository." Available: <https://www.who.int/data/gho>
5. Bray, F., et al. (2018). "Global cancer statistics 2018: GLOBOCAN estimates of incidence and mortality worldwide for 36 cancers in 185 countries." *CA: A Cancer Journal for Clinicians*, 68(6), 394-424.

Required Primary Research Sources:

CRITICAL: The following sources must be contacted for actual data validation:

Healthcare Cost Data:

- Aga Khan University Hospital - Patient billing and treatment cost databases
- Shaukat Khanum Memorial Cancer Hospital - Annual cost reports and patient data
- Combined Military Hospital system - Public sector treatment costs
- Pakistan Health Research Council - National healthcare cost studies

Diagnostic Pricing:

- Pakistan Society of Radiology - Current imaging fee schedules
- Major diagnostic laboratories (Chughtai Labs, Excel Labs, IDO) - Current pricing
- Provincial health departments - Public sector diagnostic costs

Policy and Regulatory:

- Ministry of National Health Services - Digital health implementation guidelines
- Pakistan Medical and Dental Council - Technology adoption regulations
- National Information Technology Board - Healthcare digitization policies

Market Research:

- Pakistan Bureau of Statistics - Healthcare expenditure data
- State Bank of Pakistan - Healthcare sector financial reports
- Major health insurance providers - Coverage and cost analysis

Important Disclaimer

This document is a framework for investment analysis, not a guarantee of returns. All financial projections are estimates based on available international literature and require validation through comprehensive primary research with Pakistani healthcare institutions. Actual implementation should proceed with pilot programs to validate all assumptions before full-scale deployment.

The authors recommend conducting thorough due diligence with the healthcare institutions listed above before making investment decisions. This analysis serves as a starting point

for further research and should not be used as the sole basis for financial commitments.

Next Steps:

- 1. Initiate contact with listed healthcare institutions**
- 2. Conduct primary cost validation studies**
- 3. Develop pilot program proposal**
- 4. Refine financial projections based on actual data**
- 5. Proceed with phased implementation approach**